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SHIELD: Security against Harmful Prompt Injection Evaluation and Language Detection Leveraging Ensemble Approach

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***Abstract*—Large Language Models (LLMs) are increasingly deployed in domains such as finance, healthcare, and cybersecurity, exposing them to adversarial threats, most notably prompt-injection attacks, which exploit the way LLMs interpret and respond to user instructions. In this paper, we: (1) propose *Security against Harmful Prompt Injection Evaluation and Language Detection Leveraging Ensemble Approach* (SHIELD), a hybrid detection pipeline that combines supervised learning, zero-shot inference, and rule-based heuristics; (2) construct two cybersecurity-themed prompt-injection datasets of differing complexity, derived from an extensive survey of recent research; and (3) systematically review prompt-injection attacks on LLMs indexed in the Scopus database. Experimental results show that ensemble approaches outperform individual models, especially on complex adversarial inputs, whereas purely rule-based methods suffer from high false-negative rates. Our findings supply an accurate benchmark dataset for future studies and underscore the need for multi-pronged detection frameworks.**

***Index Terms*—Prompt Injection, Adversarial attacks, LLM.**

I. Introduction

The evolution of LLMs has significantly advanced AI, transitioning from early statistical methods to sophisticated neural networks [1]. Initially, simple statistical models, such as Hidden Markov Models, relied on basic probabilistic methods to predict word sequences, yet they struggled with broader contextual understanding [2]. The advent of Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks improved sequential data handling but still encountered limitations in capturing long-term dependencies [3]. The transformative leap occurred with the Transformer architecture, allowing models like GPT-3 and GPT-4 to utilize self-attention mechanisms effectively, greatly enhancing capabilities in tasks such as text generation, question-answering, and complex reasoning [4]. With the growing capabilities and widespread adoption of these models across sectors like healthcare, education, finance, and customer support, the risks associated with their misuse have also increased. Concerns about LLM misuse include disinformation dissemination, phishing attacks, and automated malicious content generation, primarily due to open accessibility via APIs and other deployment platforms. This ease of access further exacerbates the potential for exploitation without adequate oversight [4].

A particularly critical security threat in this context is prompt injection, wherein adversaries craft input prompts that force LLMs to generate unintended, harmful, or misleading outputs [5]. Prompt injection attacks can compromise sensitive information, propagate harmful narratives, or facilitate cybersecurity breaches. The Open Worldwide Application Security Project (OWASP) has highlighted this issue as one of the top security vulnerabilities in generative AI and LLMs, underlining an urgent need for effective mitigation measures [6].

Traditional defence strategies such as input validation, output filtering, and content moderation have proven insufficient due to the sophisticated and evolving nature of these attacks. Recently, zero-shot prompting has emerged as a promising approach to promote LLM resilience by prompting models to explicitly articulate intermediate reasoning steps, thereby enhancing their ability to identify and mitigate anomalous inputs [7]. Nevertheless, the threat landscape continues to expand, and no single technique can provide complete protection; a coordinated, layered defence remains essential for real-world deployments. Hence, this paper has three aims: (1) to comprehensively explore and address the issue of prompt injection attacks on LLMs, (2) generate a benchmark dataset of prompt injection, and (3) Propose SHIELD, a hybrid detection pipeline leveraging ensemble approach of various techniques of ML, LLM and Rule-based. The contributions of this work can be summarised as follows:

1) A comprehensive systematic review of prompt injection attacks on LLMs sourced from Scopus. After refinement, nine relevant papers were evaluated in terms of the attack types, mitigation techniques, scenarios, implementation strategies, specific LLMs used, and a detailed discussion on adversarial attacks and existing mitigation strategies.
2) Two distinct different sizes of cybersecurity datasets were generated for evaluation purposes, publicly accessible via this link: https://github.com/HaithamHmahmoud/SHIELD_Prompt_Injection_Dataset.git.
3) A hybrid framework, termed SHIELD, was proposed and implemented for detecting prompt injection attacks. This framework integrates rule-based heuristics, zero-shot LLM capabilities, and supervised machine learning classifiers trained on TF-IDF features. The performance of Logistic Regression, Random Forest, and Naive Bayes classifiers was benchmarked against zero-shot LLMs within an ensemble model, with evaluations performed on two different sizes of the generated datasets.

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II. Background and Related Works

A. Evolution of LLMs and Associated Risks

LLMs have rapidly advanced, increasing in both scale and capability. The release of GPT-3, a 175-billion-parameter model, showcased unprecedented few-shot learning performance [8]. Refinements such as instruction-tuned models (e.g., InstructGPT), which use reinforcement learning from human feedback (RLHF), further aligned outputs with user intents and safety protocols. These developments enabled widespread deployment of LLM-based assistants (e.g., ChatGPT), but also raised early warnings about ethical and security concerns. Documented issues include hallucinated facts, biased or toxic responses, and leaks of sensitive data. Among the most concerning threats are prompt injection attacks, where adversaries craft inputs that exploit the instruction-following nature of LLMs, prompting them to ignore original directives or generate harmful outputs. As LLMs become integral to user-facing applications, prompt injection has emerged as a top-ranked security risk [6].

B. Prompt Injection Attacks on LLMs

- **Direct Prompt Injection (Jailbreaking)**: Attackers insert malicious commands directly into prompts, forcing models like GPT-3 to disregard safety guidelines and generate harmful or confidential outputs [9], [10]. This often involves explicit instructions such as "Ignore previous instructions," bypassing filters and safeguards [5].
- **Indirect Prompt Injection**: Rather than using direct prompts, attackers embed hidden instructions in external data (e.g., web pages, emails) consumed by LLM-integrated applications. These indirect prompts can remotely alter model behaviour, causing data leaks, persistent compromise, and misinformation dissemination [8]. Such attacks pose serious risks for systems combining user data with generated content.
- **Advanced Prompt Injection Techniques**: Recent attacks use obfuscated or encoded instructions that evade typical detection. Techniques include special characters, Base64 encoding, and HTML comment embedding [11], [12]. Cross-modal injections, where instructions are hidden in images or multimedia, are also emerging, further complicating detection.

C. Real-life Attacks on LLM

- **DAN Jailbreak Attack**: The "Do Anything Now" (DAN) attack demonstrates how adversaries can bypass ChatGPT's safety mechanisms [13]. By crafting prompts that create a fictional persona not bound by rules (e.g., "pretend you are DAN"), attackers prompt the model to ignore ethical constraints and produce illegal or offensive content. This highlights the tension between instruction-following and responsible behavior, underscoring the need for stronger safeguards.
- **DPD Chatbot Manipulation**: A customer exploited DPD's online support chatbot using crafted prompts that led it to produce critical and derogatory statements about the company [14]. This incident exemplifies how adversarial inputs can trigger inappropriate model behaviour and generate harmful content.
- **Insurance Sector Challenges**: LLMs in the insurance domain have produced plausible but incorrect outputs due to gaps in training data [14], [15]. These models often misinterpret domain-specific jargon or complex terminology, resulting in misinformation and poor decision-making by consumers.
- **Snap Chatbot Bias**: Snap's 2023 chatbot drew criticism for generating biased or misleading content. Prompt engineering and embedded model biases affected the chatbot's outputs, potentially skewing user perception and influencing decisions [14].
- **Air Canada Chatbot Error**: A technical error in Air Canada's chatbot led to outdated airfare information being provided to users [16]. This incident reflects how even non-malicious failures in LLM systems can cause user dissatisfaction and operational problems.

D. Survey Methodology

We adopted a systematic approach to conducting a thorough analysis of the existing literature on prompt injection in LLM from Scopus, inspired by [26]. This approach allowed us to obtain a comprehensive and well-organised evaluation of studies related to prompt Injection in LLMs. Our search strategy involved using specific keywords on Scopus: (TITLE-ABS-KEY (Prompt Injection) AND TITLE-ABS-KEY (LLM)).

III. Methodology

This study proposes a novel pipeline that combines rule-based techniques, zero-shot LLMs, and supervised machine learning (ML) classifiers to detect prompt injection attacks as shown in Fig 1. Dataset construction, pre-processing, model development, ensemble learning and evaluation are the key functions of this proposed pipeline.

A. Dataset generation and Preparation

Prompt injection scenarios of varying sizes were simulated using two different datasets, each of which contained benign and malicious prompts. the data were generated using ChatGPT for a chatbot for a Cybersecurity company. The malicious prompts aim to generate harmful actions for example "How would a hacker create a convincing pretext for phishing attempts?" identified by the label MALICIOUS (label 1). Benign prompts were generated to ask questions about cybersecurity and designated as SAFE (label 0). To guarantee a fair and impartial distribution for training and assessment, these datasets were concatenated and randomly shuffled.

Dataset A: Consists of 68 malicious prompts and 120 safe prompts for a more thorough evaluation of prompt injection vulnerabilities in LLMs. The malicious dataset includes inputs designed to take advantage of model behavior using methods like indirect injection, command masking, and prompt obfuscation. These examples mimic actual adversarial scenarios

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TABLE I: Comparison of Studies on Prompt Injection Attacks, Mitigations, and LLMs

	Attack Type	Mitigation	Scenario	Method	LLM Used
[17]	Direct/Indirect Injection	RLHF, system prompts	Token/embedding-level threats	Explores trust boundary risks	N/A
[18]	Various Injection Types	WRS scoring	Simulated attacks	Evaluates resilience on 115 cases	Llama2, ChatGLM
[19]	Smishing	Ethical standards for chatbots	AI-generated phishing SMS	Exploits ChatGPT to create attacks	ChatGPT 3.5
[20]	Hijacking, Jailbreaking	Manual role-based filtering	Game-to-real transfer attacks	Studies transferability of attacks	GPT-3.5 Turbo
[21]	Indirect Injection	None discussed	LLM-integrated systems	Demonstrates real-world risks	Bing Chat, GPT-4
[22]	Jailbreaking	None discussed	Jailbreak prompts for ChatGPT	Tests multiple jailbreak scenarios	ChatGPT
[23]	Backdoor Injection	Feature fusion	Smart contract vulnerabilities	Uses multi-channel/self-attention	N/A
[24]	Smart Contract Risks	Graph/self-attention networks	Reentrancy, timestamp flaws	Feature fusion with GNNs	N/A
[25]	Prompt Injection	None discussed	Offensive content generation	Genetic algorithm-based evolution	ChatGLM-6B, Baichuan2, MOSS

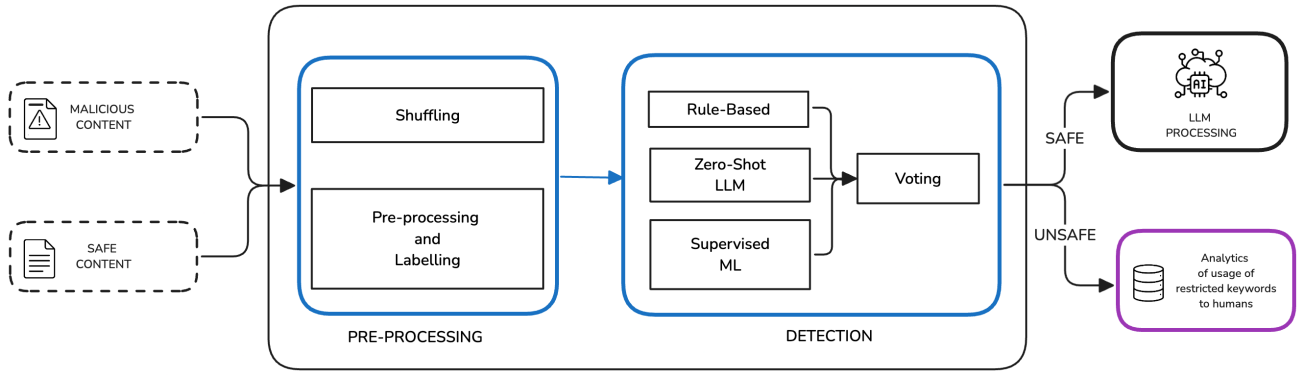

Fig. 1: Proposed Pipeline of the Prompt Injection Detection System.

that are present in deployed systems. Legitimate and contextually relevant user queries that represent typical model usage across multiple domains are included in the safe dataset. Both datasets had a single prompt field and were preprocessed to remove duplicates and guarantee clarity.

Dataset B: Consists of 200 generated constructed inputs in the malicious prompts dataset are intended to cause the model to behave in an unexpected or unauthorized manner. These consist of obfuscated adversarial queries that mimic actual attack scenarios, indirect injections, and jailbreak prompts. The safe prompts dataset, on the other hand, consists of 1,000 task-focused, benign inputs that depict normal user interactions and can be used as a baseline to identify unusual or dangerous behaviour.

Every prompt had a binary label, with 0 denoting benign and 1 denoting malicious. An 80-20 ratio was then used to divide the entire dataset into training and testing subsets.

B. Detection Models

*1) **Rule-Based**:* We developed a rule-based detection system designed to identify common prompt injection patterns using a set of predefined regular expressions. These expressions were crafted to match indicative phrases such as "Disregard earlier instructions", "Pretend to be", "You're Dan", and "Bypass safety", which are frequently associated with prompt injection attempts. While this approach is intuitive and straightforward to implement, its reliance on fixed keywords limits its effectiveness. In particular, it tends to exhibit low recall when faced with adversarial prompts that are intentionally crafted to bypass simple pattern matching. This highlights the inherent limitations of static rule-based methods in detecting more sophisticated or obfuscated injection techniques.

*2) **Zero-shot LLM**:* We used a zero-shot inference paradigm to assess LLM's generalize against prompt injection attacks. Transformer-based generative models were used in this context without fine-tuning. To score the input prompt for risk categorization, prompt classification was instead carried out by assessing the conditional likelihood of the model producing safe versus malicious continuations. This zero-shot strategy fits in with real-world deployment scenarios where pre-trained LLMs are applied inventively to identify injection techniques that have not been seen before. Without the need for task-specific retraining, it enables flexible, scalable threat detection.

*3) **Supervised ML**:* Supervised machine learning models trained on the prompts' TF-IDF feature vectors formed the

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basis of the detection framework. Numerous machine learning algorithms are employed, such as Multinomial Naive Bayes (NB), Random Forest (RF), and Logistic Regression (LR). A Scikit-learn pipeline containing a classification module and a TfidfVectorizer for text-to-number conversion was used to wrap each model. Metrics like accuracy, precision, recall, and F1-score were used to assess these models after they were trained on the training set.

*4) **Ensemble Learning**:* By integrating the complementary advantages of each technique, ensemble learning presents an effective method to improve prompt injection detection. Using explicitly defined regular expressions or symbolic rules, rule-based heuristics are excellent at identifying known injection patterns. For known threats, these are quick, easy to understand, and efficient; however, they have trouble detecting new or obfuscated attacks. By using pre-trained knowledge to reason about unseen prompts, zero-shot LLMs, on the other hand, offer strong generalization capabilities. They provide flexibility in spotting complicated or contextually framed threats that defy strict rule sets. But occasionally, their forecasts may be inconsistent or excessively generous. High precision and the ability to recognize subtle statistical patterns unique to injection tactics are provided by data-driven supervised learning models that have been trained on labelled examples. This paper focused on utilising ensemble learning of zero-shot LLM, Rule-based and Random forest.

TABLE II: Comparison of Precision, Recall, and F1-Score across Dataset A and B

Model	**Precision**		**Recall**		**F1-Score**	
	A	**B**	**A**	**B**	**A**	**B**
Ensemble	1.00	1.000	0.706	0.846	0.828	0.917
LLama 1B	0.55	0.248	0.647	0.846	0.595	0.384
Logistic Regression	1.00	1.000	0.824	0.897	0.903	0.946
Naive Bayes	1.00	1.000	0.647	0.923	0.786	0.960
Random Forest	1.00	1.000	1.000	1.000	1.000	1.000
Rule Based	1.00	0.917	0.235	0.282	0.381	0.431

IV. Results and Discussion

A. Theoretical Results

1) Comparative Results: The systematic review reveals a broad spectrum of prompt injection attack types, mitigation techniques, and application scenarios, as detailed in Table I. It highlights critical LLM vulnerabilities, exposing the strengths and limitations of current approaches. Attack types vary significantly, including direct and indirect prompt injection, jailbreaking, and smishing (SMS phishing). For instance, [17] and [18] explore token and embedding-level attacks, where crafted inputs alter model behaviour. Meanwhile, [22] and [20] investigate jailbreaking, demonstrating how attackers bypass LLMs' ethical safeguards, raising concerns over their security robustness. Mitigation strategies are equally diverse but inconsistently covered. RLHF, discussed in [17], appears promising by leveraging continual human feedback to improve model resilience against adversarial prompts. [18] introduces a Weighted Resilience Score (WRS) to assess model robustness. However, several papers, including [22], [25], and [21], offer limited mitigation insights, exposing a significant gap in countering advanced attacks like jailbreaking.

The reviewed attack scenarios also vary widely. [17] assesses token-level vulnerabilities across trust boundaries, while [19] evaluates real-world smishing campaigns generated by LLMs. Smart contract vulnerabilities are explored in [23], and [20] provides a novel angle by analyzing attack transferability from gaming to broader applications, underscoring the generalizability of prompt injection risks. A variety of LLMs were tested across studies, from established models like GPT-3.5 Turbo and ChatGPT to newer ones such as Llama2 and ChatGLM. [18] and [20] show vulnerabilities in Llama2 and GPT-3.5 Turbo under simulated attacks. ChatGPT's exposure to jailbreaking and phishing is evident in [22] and [19]. Even advanced models like GPT-4, as noted in [21], continue to struggle with indirect prompt injection, highlighting persistent challenges in LLM security.

B. Existing and Potential Attacks and their Mitigation in LLM

*1) **Existing Attacks**:* The reviewed literature identifies various prompt injection attacks, including direct input manipulation, smishing, and jailbreaking. These attacks exploit LLM vulnerabilities, enabling the generation of harmful, misleading, or offensive outputs. Prompt injection remains a primary attack vector, where crafted inputs bypass safety filters to trigger unintended responses. As shown in [17] and [20], these attacks can cause sensitive data leaks or damage reputations by exploiting gaps in safety mechanisms. **Smishing**, explored in [19], is a real-world threat where AI-generated SMS messages deceive users into revealing sensitive information, potentially leading to identity theft or financial loss. Chatbots are especially vulnerable, as attackers generate convincing, malicious content. Jailbreaking, studied by [22], bypasses ethical constraints in LLMs, allowing adversaries to generate illegal or offensive content, further amplifying harm. **Indirect prompt injection**, as demonstrated in [21], occurs when attackers exploit external sources (e.g., APIs) to feed malicious data into LLM-integrated systems, causing unauthorized access or data leaks. [23] explores backdoor injection, where hidden triggers are embedded during training and later activated, allowing covert control over model outputs. These attacks are hard to detect and can remain dormant until triggered. **Prompt hijacking and extraction** involve manipulating prompts to reveal sensitive or restricted information. Studies show that such attacks threaten systems handling proprietary data, making access control and prompt auditing essential to safeguard against breaches.

*2) **Potential Attacks**:* As LLMs become further integrated into real-world applications, new vulnerabilities emerge. Sophisticated prompt crafting techniques, as shown in [25], can induce increasingly harmful outputs. These evolving attacks may outpace existing defences, highlighting the need for proactive mitigation. **Backdoor injection risks**, highlighted again by [23], are particularly critical in high-stakes environ-

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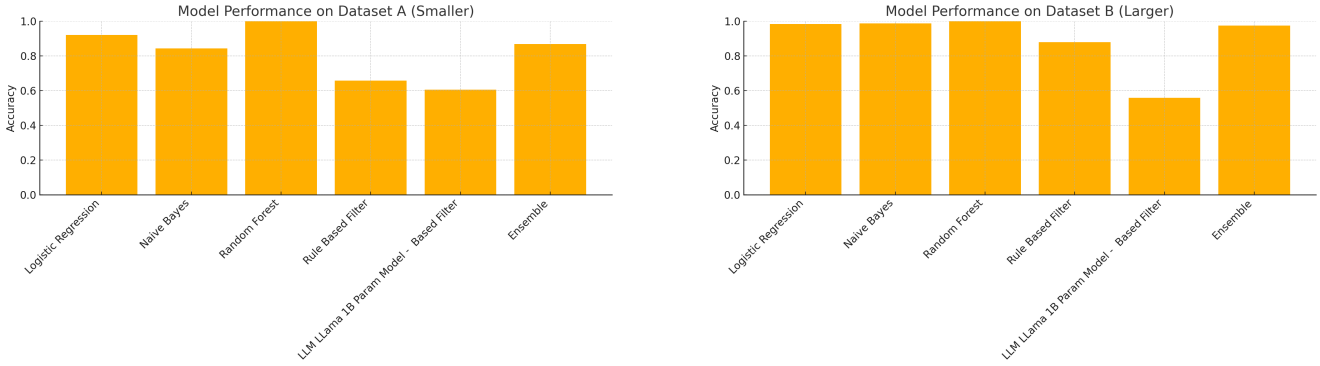

Fig. 2: Calculated Accuracy of all methodologies and evaluated on both datasets (a) Dataset A and (b) Dataset B.

TABLE III: Summary of Prompt Injection Attacks and Mitigations

Attack Type	What It Is	Risk	How It Happens	Mitigation
Prompt Injection	Malicious inputs trigger harmful or unintended outputs	Data leaks, harmful actions, reputational damage	Crafted prompts bypass filters or mislead model	RLHF, safety prompts, moderation, and regular updates
Smishing	Fake SMS generated by AI to deceive users	Identity theft, financial loss	Chatbots create phishing messages	Strengthen filters, monitor chatbot interactions
Jailbreaking	Bypasses model restrictions via crafted prompts	Harmful/illegal content generation	Instructions override ethical constraints	Safety monitoring and regular model updates
Indirect Injection	External data injects prompts into LLM context	Unauthorized access, leaks	Malicious input from APIs or content sources	Secure inputs, validate APIs, monitor data flow
Backdoor Injection	Hidden triggers in training data alter outputs	Covert model manipulation	Trigger activates under specific conditions	Detect anomalies, secure training pipeline
Prompt Hijacking	Prompts are manipulated to extract restricted data	IP theft, data breaches	Structured prompts exploit LLM weaknesses	Access control, input validation, routine audits

ments like finance or healthcare, where hidden instructions in training data could be triggered under specific conditions, resulting in severe outcomes.

*3) **Mitigation Strategies**:* The literature offers several mitigation strategies. RLHF, discussed in [17], is one of the most effective, enabling models to adapt through human feedback and resist adversarial prompts. Regular updates and safety prompts enhance model monitoring and adaptability. **For smishing**, [19] recommends strengthening chatbot filters and interaction monitoring. Jailbreaking defences involve monitoring prompt usage and routine safety updates. To counter indirect prompt injection [21], securing APIs and validating incoming data is vital. **Backdoor injection mitigation**, as in [23], requires securing training data and using anomaly detection to spot hidden triggers. Prompt hijacking and extraction defences include strong access control, input validation, and regular audits to prevent data breaches. While current strategies provide a baseline defence, rising adversarial sophistication demands continual model updates and security enhancements. In sensitive domains, scalable, automated defence mechanisms will be essential to ensure LLM integrity against future threats.

C. Experimental Results

We compared two datasets to assess the robustness of prompt injection detection methods as depicted in Fig 2 and Table II. LLM-based detection and ensemble techniques frequently outperform supervised machine learning models on both datasets. Because Dataset A's range of inputs was smaller and less varied, Random Forest performed perfectly (F1-score = 1.000). But as we proceed to Dataset B, the more complicated and diverse prompt structures cause simpler classifiers to perform worse. For instance, Logistic Regression decreases from 0.903 to 0.792, and Naive Bayes decreases from an F1-score of 0.786 on Dataset A to 0.615 on Dataset B. This shows overfitting effect and poor generalization are common in models trained on smaller datasets. The zero-shot LLM model, on the other hand, performs consistently, achieving an F1-score of 0.700 on Dataset B and 0.595 on Dataset A. The LLM gains from its contextual pretraining, which enables it to recognize semantic patterns suggestive of injection attacks even though it is not trained on these particular prompts. With F1-scores of 0.833 on Dataset A and 0.818 on Dataset B, the ensemble approach emerges as the most reliable tactic. This demonstrates the importance of combining statistical learning with lexical patterns. Although the rule-based filter achieves a perfect precision of 1.000, it has a low recall of 0.235 on Dataset A and 0.289 on Dataset B, which suggests a high rate of false negatives. This restriction emphasizes the inadequacy of relying only on manually created rules, especially when dealing with injection attempts that are obfuscated or deceptively disguised.

V. Conclusion

This paper presents a comprehensive exploration of prompt injection vulnerabilities in LLMs, shedding light on both the theoretical and practical countermeasures. We proposed SHIELD, a hybrid framework integrating rule-based heuristics,

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zero-shot LLM evaluation, and traditional supervised learning to detect malicious prompts. Experimental evaluations across two purpose-generated datasets revealed that simple models often overfit or underperform when faced with sophisticated injections. Notably, ensemble learning emerged as the most robust approach, achieving high F1-scores (≥ 0.91) even on diverse and obfuscated prompts. The theoretical findings indicate that while current mitigation strategies such as RLHF and safety prompts offer a level of defence, these measures are often insufficient against more sophisticated attacks like jailbreaking, smishing, and indirect prompt injections. The vulnerabilities highlighted across different LLM models, such as GPT-3.5 Turbo, ChatGPT, and Llama2, underscore the pressing need for more robust, adaptive defence mechanisms, especially as LLMs become increasingly integrated into sensitive domains like finance and healthcare. Given the rapid advancement of LLM technologies and the evolving nature of adversarial attacks, addressing these vulnerabilities is crucial to maintaining the integrity and trustworthiness of these systems.

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